STATISTICS SURVIVAL WALL

CORE FOUNDATIONS & THEORIES

GREEK SYMBOLS & NOTATION

μ	Population Mean
$\bar{x}$	Sample Mean
σ	Population Std Dev
s	Sample Std Dev
α	Alpha (Type I Error Rate)
β	Beta (Type II Error Rate)
χ^2	Chi-Squared Statistic
F	F-Statistic (ANOVA/Reg)
r	Pearson Correlation
R^2	Variance Explained
η^2	Eta Squared (Effect)
d	Cohen's d (Effect)
n	Sample Size
df	Degrees of Freedom

DESCRIPTIVE STATS

- **Mean:** Average. Sensitive to outliers.
- **Median:** Middle value (50th percentile). Robust.
- **Mode:** Most frequent value.
- **Variance (s^2):** Avg squared deviation from mean.
- **Std Dev (s):** Square root of variance. In original units.
- **IQR:** Q3 - Q1. Middle 50% of data.

DISTRIBUTION CONCEPTS

- **Population:** The entire group (μ, σ).
- **Sample:** Subset ($\bar{x}$, s).
- **Sampling Dist:** Dist of a statistic (e.g., means) across infinite samples.

CENTRAL LIMIT THEOREM

As sample size ($n \geq 30$) increases, the sampling distribution of the mean becomes normally distributed, regardless of the population's shape.

NORMAL DISTRIBUTION

- Symmetrical, bell-shaped.
- **68-95-99.7 Rule:** % of data within 1, 2, and 3 standard deviations.

CONFIDENCE INTERVALS (CI)

Range containing the true parameter with X% confidence.

Level	Z-Crit	Interpretation
90%	1.645	Narrow, higher error
95%	1.960	Standard baseline
99%	2.576	Wide, conservative

ALPHA, BETA & POWER

Decision	H0 True	H0 False
Reject H0	Type I Error (α)	True Pos (Power)
Fail Reject	True Negative	Type II Error (β)

- **α (Alpha):** Prob of rejecting a true H0 (False Positive).
- **β (Beta):** Prob of failing to reject false H0 (False Negative).
- **Power ($1 - \beta$):** Prob of correctly rejecting a false H0.

HYPOTHESIS WORKFLOW

1. Formulate Question
↓
2. State H0 & HA
↓
3. Choose Alpha (α)
↓
4. Check Assumptions
↓
5. Choose & Run Test
↓
6. Interpret p-value
↓
7. Check Effect Size

ANATOMY OF A STATISTICAL OUTPUT

Most software prints a cryptic block. Here is how to decode the matrix.

```
> t.test(Group_A, Group_B, var.equal=FALSE)

    Welch Two Sample t-test

data: Group_A and Group_B
t = 2.841, df = 47.3, p-value = 0.0067
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 1.3225  7.1031
sample estimates:
mean of x  mean of y
 126.33     122.11
```

t (Test Statistic)
The calculated signal-to-noise ratio. Larger absolute value = stronger signal.

df (Degrees of Freedom)
Sample size adjusted for parameters. With Welch, this gets fractional (47.3).

p-value (Decision)
Probability of data assuming H0 is true. If $p < \alpha$ (e.g. 0.05), Reject H0.

CI (Plausible Range)
If the CI does NOT cross 0, the difference is statistically significant.

STATISTICAL MANTRAS

STANDARD RULE:

**IF THE P IS LOW,
THE NULL (H0) MUST GO!**

MASTER CUSTOM RULE:

**IF THE P IS HIGH,
THE 1 (H1) MUST DIE! 🍌**

- **Fail to reject H0 ≠ Accept H0.** (Lack of evidence is not proof of absence).
- **Significant ≠ Important.** (Check Effect Size!)
- **Correlation ≠ Causation.**

TEST SELECTION PROTOCOL & CORE TESTS

TEST SELECTION TERMINAL

```
WHAT IS YOUR OUTCOME VARIABLE?
 ├── NUMERICAL (Continuous/Interval)
 │   ├── Compare to a fixed target value ──────────────> One Sample t-Test
 │   ├── Compare groups (Categorical Predictor)
 │   │   ├── 2 Independent Groups ─────────────────────> Welch Two-Sample t
 │   │   ├── 2 Related/Paired Groups (Before/After) ───> Paired t-Test
 │   │   └── 3+ Independent Groups ────────────────────> One-Way ANOVA
 │   └── Relationship (Numerical Predictor)
 │       ├── Linear Relationship ──────────────────────> Pearson Correlation
 │       └── Monotonic (Ranked) Relationship ──────────> Spearman Correlation
 └── CATEGORICAL (Nominal/Ordinal)
     ├── Proportion (1 Variable, Binary Outcome) ──────> Binomial Test
     ├── Frequencies fit expected distribution? ───────> Chi² Goodness of Fit
     └── Association between 2 Variables ──────────────> Chi² Independence
```

ASSUMPTION FAILURE MATRIX

Parametric tests require normality and equal variance. If assumptions fail, use non-parametric equivalents (Rank-based).

Wanted Test	Broken Assumption	Use Instead
Indep. t-Test	Unequal Variance	**Welch t-Test**
Indep. t-Test	Non-Normal (Outliers)	**Mann-Whitney U**
Paired t-Test	Non-Normal	**Wilcoxon Signed Rank**
ANOVA	Non-Normal	**Kruskal-Wallis**
Pearson (r)	Non-Normal / Non-Linear	**Spearman (ρ)**

1. WELCH TWO-SAMPLE T

H0: μ1 = μ2.
Compares means of 2 independent groups. Doesn't assume equal variance (Standard!).

PYTHON
```
stats.ttest_ind(a, b, equal_var=False)
```
R
```
t.test(val ~ grp, var.equal=FALSE)
```

2. PAIRED T-TEST

H0: μ_diff = 0.
Compares means from same group at different times (Before vs After).

PYTHON
```
stats.ttest_rel(before, after)
```
R
```
t.test(before, after, paired=TRUE)
```

3. ONE-WAY ANOVA

H0: μ1 = μ2 = μ3...
Compares means of 3+ groups. If p<0.05, at least one group differs. (Needs Post-Hoc!).

PYTHON
```
stats.f_oneway(g1, g2, g3)
```
R
```
summary(aov(val ~ grp, data=df))
```

4. CHI-SQUARE INDEP.

H0: Variables are independent.
Tests association between 2 categorical variables (e.g. Pet Type x City).

PYTHON
```
chi2_contingency(pd.DataFrame(obs))
```
R
```
chisq.test(table_data)
```

5. PEARSON CORRELATION

H0: r = 0.
Measures linear strength between 2 continuous variables (-1 to 1).

PYTHON
```
stats.pearsonr(df['x'], df['y'])
```
R
```
cor.test(df$x, df$y)
```

6. MANN-WHITNEY U

H0: Distributions equal.
Non-parametric alternative to independent t-test (compares ranks, not means).

PYTHON
```
stats.mannwhitneyu(grp_a, grp_b)
```
R
```
wilcox.test(x, y, paired=FALSE)
```

7. WILCOXON SIGNED RANK

H0: Median diff = 0.
Non-parametric alternative to paired t-test.

PYTHON
```
stats.wilcoxon(before, after)
```
R
```
wilcox.test(before, after, paired=TRUE)
```

8. KRUSKAL-WALLIS

H0: Medians equal.
Non-parametric alternative to ANOVA (3+ groups).

PYTHON
```
stats.kruskal(g1, g2, g3)
```
R
```
kruskal.test(val ~ grp)
```

STATISTICS SURVIVAL WALL

REGRESSION, EFFECT SIZES & REPORTING

LINEAR & MULTIPLE REGRESSION

$$y = \beta 0 + \beta 1x1 + \beta 2x2 + ... + \varepsilon$$

Models the relationship between predictors (x) and a continuous outcome (y).

Regression Decoder:
Intercept (β0): Predicted y when all x = 0.
Coefficient (Slope β1): Expected change in y for a 1-unit increase in x, holding other x constant.
Residual (ε): Distance from data point to regression line.
R^2 (R-squared): % of variance in y explained by the model.
Adjusted R^2: R^2 penalized for adding useless predictors (always use this for Multiple Reg).

Assumptions & Diagnostics:

- **Linearity:** Plot residuals vs fitted.
- **Homoscedasticity:** Equal variance of residuals.
- **Normality:** Q-Q Plot of residuals.
- **Multicollinearity:** Check VIF (Variance Inflation Factor). If VIF > 5, predictors are highly correlated.

PYTHON
```
import statsmodels.formula.api as smf
smf.ols("y ~ x1 + x2", data=df).fit().summary()
```
R
```
summary(lm(y ~ x1 + x2, data=df))
```

LOGISTIC REGRESSION

Predicts a **Binary** outcome (0/1, Yes/No).

- **Odds:** P(Event) / P(No Event).
- **Logit:** Natural log of the odds. The model runs linearly on logits.
- **Odds Ratio (OR):** e^(Coef). OR > 1 → Increases likelihood OR = 1 → No effect OR < 1 → Decreases likelihood

PYTHON
```
smf.logit("y ~ x", data=df).fit()
```
R
```
glm(y~x, family=binomial)
```

POST-HOC ANALYSIS

ANOVA answers: "Is there ANY difference?"
Tukey answers: "WHICH groups differ?"

- **Tukey HSD:** Compares all pairs safely.
- **Bonferroni:** p_new = p * (num_tests). Strict!

PYTHON (TUKEY)
```
pairwise_tukeyhsd(df['y'], df['grp'])
```

EFFECT SIZES (PRACTICAL SIGNIFICANCE)

STATISTICAL SIGNIFICANCE ≠ PRACTICAL SIGNIFICANCE

A p-value only tells you an effect exists. Effect size tells you if you should care.

Cohen's d (t-Test)	Pearson r (Corr)	Eta² η^2 (ANOVA)
0.2 = Small	0.1 = Weak	0.01 = Small
0.5 = Medium	0.3 = Moderate	0.06 = Medium
0.8 = Large	0.5 = Strong	0.14 = Large

REPORTING CHECKLIST

Always report these elements in APA format:

✓ Test used
✓ Test statistic (t, F, χ^2)
✓ Degrees of freedom (df)
✓ Exact p-value (p = 0.034)
✓ Effect size (d, η^2, OR)
✓ Confidence interval (CI)

PYTHON ↔ R TRANSLATION

Concept	Python	R
Mean	df.mean()	mean(x)
SD	df.std()	sd(x)
Summary	df.describe()	summary(x)
Crosstab	pd.crosstab()	table()
Plot	seaborn	ggplot2

STATISTICS SURVIVAL WALL

MACHINE LEARNING & MODEL EVALUATION

DATA SPLITTING & VALIDATION

Never test a model on the data it trained on (Data Leakage / Overfitting).

- **Train Set (70-80%):** Used by the algorithm to learn patterns.
- **Test Set (20-30%):** Unseen data to evaluate final performance.
- **Validation Set:** Used to tune hyperparameters.

K-FOLD CROSS VALIDATION

Splits data into 'k' chunks. Trains on k-1, tests on the 1 remaining. Repeats k times. Averages the scores. More robust than a single split.

PYTHON (SCIKIT-LEARN)

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

CLASSIFICATION: THE CONFUSION MATRIX

Compares Actual Truth vs. Model Prediction for binary classification.

		Predicted Condition	
		Pred Positive (1)	Pred Negative (0)
Actual	Pos (1)	True Positive (TP)	False Negative (FN) - Type II Error
	Neg (0)	False Positive (FP) - Type I Error	True Negative (TN)

Imbalanced Data Warning:

If 99% of emails are normal and 1% is spam, a model predicting "normal" every time has 99% accuracy but is completely useless. Do not use Accuracy for imbalanced data!

Fixes: SMOTE (Oversampling), Undersampling, Class Weights.

EVALUATION METRICS

- **Accuracy:** (TP+TN) / Total
  Overall correctness (Bad for imbalanced)
- **Precision:** TP / (TP + FP)
  When it predicts Yes, how often is it right? (Minimize False Positives)
- **Recall (Sensitivity):** TP / (TP + FN)
  Out of all actual Yes, how many did we find? (Minimize False Negatives)
- **F1-Score:** 2 * (Prec * Rec) / (Prec + Rec)
  Harmonic mean of Precision and Recall.

ROC CURVE & AUC

ROC (Receiver Operating Characteristic): Plots True Positive Rate (Recall) vs False Positive Rate across different probability thresholds.

AUC (Area Under Curve): A single metric evaluating model performance independent of threshold.

- **AUC = 0.5:** Useless (Random guessing).
- **AUC = 0.7 - 0.8:** Acceptable.
- **AUC = 0.8 - 0.9:** Excellent.
- **AUC = 1.0:** Perfect (Suspicious!).

⚠ MACHINE LEARNING TRAPS

- TRAP **Data Leakage:** Scaling or imputing missing values BEFORE train/test split. Always fit scaler on Train only!
- TRAP **Overfitting:** 99% train accuracy but 60% test accuracy. The model memorized the noise.
- TRAP **Thresholding:** `.predict()` defaults to 0.5 threshold. You might need to change it via `.predict_proba()` to optimize Recall vs Precision.